Introduction to Topology II: Homework 8

Due on March 4, 2026 at 23:59

Nick Proudfoot 13:00

Hashem A. Damrah
UO ID: 952102243

Problem 1. Suppose that $\{K_r \mid r \geq 0\}$ is a filtered complex. More precisely, each K_r is an abstract simplicial complex with the same vertex set, and $K_{r_1} \subset K_{r_2}$ whenever $r_1 \leq r_2$. Define r to be a *critical scale* if, for all $\varepsilon > 0$, $K_{r-\varepsilon} \subsetneq K_r$.

(i) Let $X = \mathbb{R}^2$ with its Euclidean metric, and let $S = \{(1,0),(0,1),(-1,0),(0,-1)\} \subset X$. Determine all of the critical scales for $\{\text{Rips}(s,r) \mid r \geq 0\}$ and describe the complex at all parameters.

(ii) With the same X and S in part (i), determine all of the critical scales for $\{\text{Cech}(S,r) \mid r \geq 0\}$ and describe the complex at all parameters.

(iii) Redo part (ii) with the same set S but this time with $X = \mathbb{S}^1$, still with the Euclidean metric.

Solution to (i). The critical scales are $r = 0, 1, \sqrt{2}$. For $0 \leq r < 1$, $\text{Rips}(S,r)$ consists of four vertices and no edges. For $1 \leq r < \sqrt{2}$, $\text{Rips}(S,r)$ consists of four vertices and four edges, but no triangles. For $r \geq \sqrt{2}$, $\text{Rips}(S,r)$ consists of four vertices, six edges, and four triangles. □

Solution to (ii). The critical scales are $r = 0, \sqrt{2}$. For $0 \leq r < \sqrt{2}$, $\text{Cech}(S,r)$ consists of four vertices and no edges. For $r \geq \sqrt{2}$, $\text{Cech}(S,r)$ consists of four vertices, six edges, and four triangles. □

Solution to (iii). The critical scales are $r = 0, \sqrt{2}$. For $0 \leq r < \sqrt{2}$, $\text{Cech}(S,r)$ consists of four vertices and no edges. For $r \geq \sqrt{2}$, $\text{Cech}(S,r)$ consists of four vertices, six edges, and four triangles. □

Problem 2. Let X and S be as in problem 1(i), and let

$$T = \left\{ \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) \right\} \subset X.$$

Compute the minimum $\varepsilon \geq 0$ such that $\{\text{Rips}(S,r) \mid r \geq 0\}$ and $\{\text{Rips}(T,r) \mid r \geq 0\}$ are ε-interleaved.

Solution. Let $X = \mathbb{R}^2$ and $S = \{(1,0),(0,1),(-1,0),(0,-1)\} \subset X$. Let T be as in the problem statement. We claim that the minimum $\varepsilon \geq 0$ such that $\{\text{Rips}(S,r) \mid r \geq 0\}$ and $\{\text{Rips}(T,r) \mid r \geq 0\}$ are ε-interleaved is $\varepsilon = \sqrt{2} - 1$. To see this, note that $\text{Rips}(S,r)$ and $\text{Rips}(T,r)$ are isomorphic for all $r < \sqrt{2} - 1$. However, at $r = \sqrt{2} - 1$, $\text{Rips}(S,r)$ consists of four vertices and four edges, while $\text{Rips}(T,r)$ consists of four vertices and no edges. Therefore, the minimum ε such that $\{\text{Rips}(S,r) \mid r \geq 0\}$ and $\{\text{Rips}(T,r) \mid r \geq 0\}$ are ε-interleaved is $\varepsilon = \sqrt{2} - 1$. □